

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2024

MA1008-1 – CALCULUS AND LAPLACE TRANSFORMS

(BT, CV, ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- If ϕ be the angle between the tangent and radius vector at any point on the curve $r = f(\theta)$ then $\tan\phi$ equals to _____
 A) $\frac{dr}{ds}$ B) $r \frac{d\theta}{ds}$
 C) $r \frac{d\theta}{dr}$ D) $\frac{d\theta}{dr}$
- The angle between the radius vector $r = a(1 - \cos\theta)$ and tangent to the vector is $\phi =$ _____
 A) $\frac{\theta}{2}$ B) θ
 C) 0 D) $\frac{\pi}{2}$
- The radius of curvature for the curve $y = f(x)$ is $\rho =$ _____
 A) $\frac{(1+y_2^2)^{3/2}}{y_1}$ B) $\frac{(1+y_1^2)^{3/2}}{y_2}$
 C) $\frac{(1+y_1^2)^{2/3}}{y_2}$ D) $\frac{(1-y_1^2)^{3/2}}{y_2}$
- The value of c obtained by applying Lagrange's MVT to the function $f(x) = x^2$ in $[0, 4]$ is
 A) 1 B) 4
 C) 3 D) 2
- If $u = x^4 + y^4 + 3x^2y^2$, then $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ is
 A) u B) $3u$
 C) $2u$ D) $4u$
- If $f(x, y, z) = x^2 + xyz + z$ then f_x at $(1, 1, 1)$
 A) 0 B) 1
 C) 3 D) -1
- If $x = r \cos \theta$, $y = r \sin \theta$ then the value of $\frac{\partial(x,y)}{\partial(r,\theta)}$ is _____
 A) r B) 0
 C) 1 D) $\frac{1}{r}$
- If $A = f_{xx}(a, b)$, $B = f_{xy}(a, b)$, $C = f_{yy}(a, b)$ then $f(x, y)$ will have a maximum at (a, b) if
 A) $f_x = 0, f_y = 0, AC < B^2$ and $A < 0$ B) $f_x = 0, f_y = 0, AC = B^2$ and $A > 0$
 C) $f_x = 0, f_y = 0, AC > B^2$ and $A > 0$ D) $f_x = 0, f_y = 0, AC > B^2$ and $A < 0$
- If $L\{f(t)\} = \bar{f}(s)$, then $L\{\sin 3t\} =$ _____
 A) $\frac{3}{s^2 - 3^2}$ B) $\frac{3}{s^2 + 3^2}$
 C) $\frac{s}{s^2 - 3^2}$ D) $\frac{s}{s^2 + 3^2}$

10. If $L\{f(t)\} = \bar{f}(s)$, then $L\left\{\frac{f(t)}{t}\right\} =$ _____
 A) $\int_s^\infty \bar{f}(s) ds$ B) $\int_0^\infty \bar{f}(s) ds$
 C) $\frac{1}{s} \bar{f}(s)$ D) $\bar{f}(s)$
11. Laplace transform of $e^{-2t} \sin 5t =$ _____
 A) $\frac{5}{(s+2)^2 + 5^2}$ B) $\frac{(s+2)}{(s+2)^2 - 5^2}$
 C) $\frac{5}{(s-2)^2 + 5^2}$ D) $\frac{2}{(s-3)^2 - 2^2}$
12. $L\{u(t-4)\} =$ _____
 A) $\frac{e^{-4s}}{s^2}$ B) e^{4s}
 C) $\frac{e^{4s}}{s}$ D) $\frac{e^{-4s}}{s}$
13. Inverse Laplace transform of $(s+2)^{-2}$ is _____
 A) te^{-2t} B) te^{2t}
 C) e^{2t} D) e^{-2t}
14. Inverse Laplace Transform of $\log\left(\frac{s+3}{s+4}\right)$ is _____
 A) $\frac{1}{t}[e^{-3t} - e^{-4t}]$ B) $-\frac{1}{t}[e^{-3t} - e^{-4t}]$
 C) $-\frac{1}{t}[e^{-3t} + e^{-4t}]$ D) $[e^{-3t} - e^{-4t}]$
15. $L^{-1}\{e^{-as} \bar{f}(s)\} =$ _____
 A) $f(t) u(t-a)$ B) $f(t-a) u(t)$
 C) $f(t-a) u(t-a)$ D) $f(t+a) u(t+a)$
16. If $f'(t)$ is continuous and $L\{f(t)\} = \bar{f}(s)$ then $L\{f'(t)\} =$ _____
 A) $\bar{f}(s) - sf(0)$ B) $s\bar{f}(s) - f(0)$
 C) $s^2\bar{f}(s) - sf(0) - f'(0)$ D) $\bar{f}(s) - sf(0) - f'(0)$
17. $\cos(5x + 6y) \frac{\partial^3 z}{\partial y^3} + \frac{\partial^2 z}{\partial x^2} - xy = 0$ is a partial differential equation of degree _____
 A) 2 B) 1
 C) 0 D) 3
18. One dimensional wave equation is _____
 A) $u_t = c^2 u_{xx}$ B) $u_{tt} = c^2 u_x$
 C) $u_t = c^2 u_x$ D) $u_{tt} = c^2 u_{xx}$
19. Solution of the partial differential equation $\frac{\partial z}{\partial x} + \cos(3x - 2y) = 0$ by direct integration is _____
 A) $z + \frac{\sin(3x - 2y)}{3} = x f_1(y)$ B) $z - \frac{\sin(3x - 2y)}{3} = f_1(y)$
 C) $z + \frac{\sin(3x - 2y)}{3} = f_1(y)$ D) $z - \frac{\sin(3x - 2y)}{3} = f_1(x)$
20. The partial differential equation obtained by eliminating arbitrary constants from the equation $z = (x-a)^2 + (y-b)^2$ is _____
 A) $2z = p^2 x + qy$ B) $2z = p^2 - q^2$
 C) $4z = p^2 + q^2$ D) $z = px - 2qy$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	State and prove Cauchy's mean value theorem.	5	L*1	1	1
b)	Prove that radius of curvature of a circle is constant.	5	L2	1	1
c)	If $u = \frac{e^{x+y}}{e^x + e^y}$ prove that $u_x + u_y = u$.	6	L2	2	1
2. a)	Find the extreme values of the function $f(x, y) = x^2 + y^2 + 6x - 12$.	5	L2	2	1
b)	Using Lagrange's mean value theorem determine c for the function: $f(x) = x^3 - 2x^2$ in $[2, 5]$.	5	L2	1	1
c)	Using Taylor's series expansion expand the function upto second degree terms $f(x, y) = \sin(xy)$ about the point $(1, \frac{\pi}{2})$.	6	L1	2	1
3. a)	With the usual notation prove that $\tan\phi = r \frac{d\theta}{dr}$.	5	L1	1	1
b)	Find the angle between the two curves $r = a(1 - \cos\theta)$ and $r = 2a \cos\theta$.	5	L1	1	1
c)	If $x = r \cos\theta$ and $y = r \sin\theta$ prove that $J J' = 1$.	6	L2	2	1
Unit – II					
4. a)	Find $L[e^{5t} \cos 5t \cdot \cos 3t]$.	5	L1	3	1
b)	Rewrite the following function using unit step function and find its L.T. $f(t) = \begin{cases} \sin t, & 0 < t \leq \frac{\pi}{2} \\ \cos t, & t > \frac{\pi}{2} \end{cases}$	5	L2	3	1
c)	Find $L^{-1} \left[\frac{7s+4}{4s^2+4s+9} \right]$.	6	L1	4	1
5. a)	Find Inverse Laplace transforms of $\log \left[\frac{s+a}{s+b} \right]$.	5	L1	4	1
b)	If $f(t)$ is a periodic function with period T such that $f(t+T) = f(t)$ for all values of t , prove that $L\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$.	5	L1	3	1
c)	Find the Laplace Transforms of $\int_0^t \frac{\cos at - \cos bt}{t} dt$.	6	L2	3	1
6. a)	Using the convolution theorem find $L^{-1} \left[\frac{1}{s^2(s-1)} \right]$	5	L1	4	1
b)	Solve the differential equation by Laplace transform method: $x''(t) + 4x(t) = 2t - 8; x(0) = 1, x'(0) = 0$.	5	L2	4	1
c)	Find $L\{t \sin^2 3t + e^{(3t+7)}\}$.	6	L1	3	1
Unit – III					
7. a)	Form a Partial differential equation by eliminating arbitrary constants from $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$.	5	L1	5	1
b)	Solve the following higher order partial differential equation by direct integration method $\frac{\partial^3 z}{\partial x^2 \partial y} + 18xy^2 + \sin(2x - 3y) = 0$.	5	L1	5	1
c)	Derive one dimensional wave equation in the form $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$.	6	L2	5	1
8. a)	Form partial differential equations by eliminating the arbitrary function from the equation $f(x^2 + 2yz, y^2 + 2zx) = 0$.	5	L1	5	1
b)	Solve the partial differential equation $\frac{\partial^2 z}{\partial y^2} - z = 0$ given $y = 0, z = e^x$ and $\frac{\partial z}{\partial y} = e^{-x}$.	5	L1	5	1
c)	Solve the following linear partial differential equation by Lagrange's method $xp + yq = z$.	6	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
